

Change Log:

04-09-2021 More examples and extend the deadline

Project Assignment

Phase 1: Business domain study and database design

Project Description

This project gives you an opportunity to practice data modeling process in designing the database for the business process/application of your group's interest.

Thing to do:

1. Select a **business process/application** of your group's interest for the project. *If you prefer to choose a different domain, please get your instructor's approval first.* You have to describe the responsibility and activities of the selected business processes.

Example of a business process: the *bank-to-bank transfer* process in the SCB bank has responsibility for handling clients' requests to transfer money from their bank account to their own or other bank accounts. The clients can request the services via a variety of channels such as mobile application, counter service, and ATM

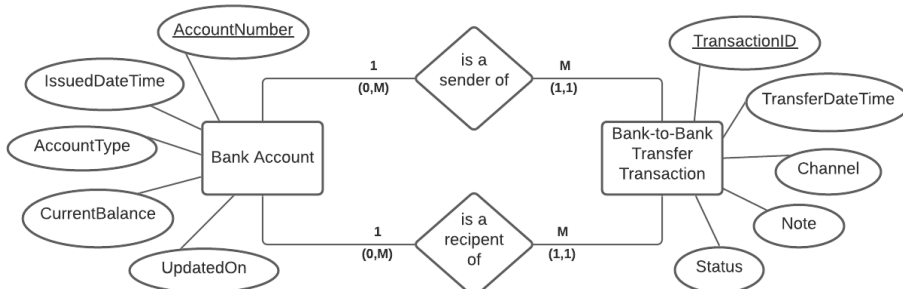
2. Study the company's business policy and and try to understand their business process so that you can extract the (databae) **business rules and constraints**. Your group should extract the requirements from documentation obtained from the company and/or their web/mobile applications. For example, you may open the SCB bank mobile application and go to the transfer money between account page. On this page, you may see that it requires a sender's bank account number and a recipient's bank account number, an amount to transfer, time to transfer (right away or schedule for later), and etc. For each request, a client can transfer money as long as the amount is within the limit and there is enough money in the sender's account (*Please note that the business requirements should be the in-depth analysis rather than the broad analysis*)

Example of business rules: Each bank-to-bank transfer requires *one* sender's bank account info and *one* recipient's bank account info. Each bank account can be used in *many* bank-to-bank transfers.

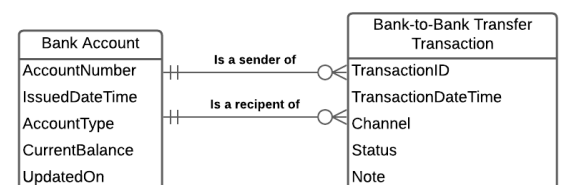
Example of constraints: The minimum and maximum amount of money in each bank-to-bank transfer is between 1 to 1,000,000 Baht.

3. Based on all business rules in step 2, your group has to design the database using **Entity Relationship Diagram (ERD)**. In this project, only **Chen Notation (preferable)** OR **Crow's Foot Notation** are allowed. Here is an example ERD from business rules in step 2. (*You have to do only ONE diagram*)

Chen Notation



Crow's Foot Notation



These diagrams are made using Lcidchart App

ITCS 241: Database Management System (3 Credits)
Semester 1/2012 Faculty of ICT, Mahidol University**Deliverables and Timeline:**

Week 5: The first report (at least 5 pages) must include the following topics:

- A business domain of your choice and a real company in that domain that you will study.
- The overview of the company including their history, and their goals (vision).
DO NOT COPY FROM THE INTERNET. Please summarize their contents in your own words.
- For the business process of your choice, describe its responsibilities, and its main activities.
- The business rules and constraints of the application/system.
 - For a group of 4 members: at least 32 business rules & 8 constraints
 - For a group of 3 members: at least 24 business rules & 6 constraints.
- The picture (e.g., photo, screen capture) of the actual forms obtained from the company used for identifying the business rules. If it is not available, you may use example from other businesses, or some templates or your own design.
- References
- Name your report as: ITCS241_secX_grY_report1_IDxxx_xxx_xxx_xxx.pdf
- **Due date: All Sections – ~~Friday 10 September, 2020~~ Sunday 12 September 2021 at 11:55 PM BKK Time**

Week 8: The second report (at least 10 pages) must include the following topics:

- The revised version of the first report (submitted in week 5).
- The **ER diagrams (ERD)** to show a blueprint of your database
 - For group of 4 members: at least 15 entities and at least 4 out of 15 entities must have at least 10 attributes
 - For group of 3 members: at least 12 entities and at least 3 out of 12 entities must have at least 10 attributes
- References
- Name your report as: ITCS241_secX_grY_final1_IDxxx_xxx_xxx_xxx.pdf
- **Due date: All Sections– ~~Friday 01 October, 2020~~ Sunday 03 October 2021 at 11:55 PM BKK Time**

***** Your diagram must be readable in the report, otherwise ZERO mark will be given.***

Note that your instructor will give you some valuable feedback based on your work. You have to revise and redesign your project according to these feedbacks.

Concerned Issues:

1. Plagiarism is NOT allowed for any cases.
2. Creativity in design and presentation is preferable.
3. Please keep in mind that you are acting as the database designer and your users are non-technical staffs of the company. Before asking them for the documentations or business forms, please be polite and you may act as their customer who wants to do their business functions.

Marking Criteria (50 points) -> This will be converted to 12% of your total grade

Topic	0	1	2	3	4	5	Max	Multiplier	Full Score
Business Domain	Provided details, but not clear	Include business name and description but lack of detail information (too general)	Include business name, description and detail information including some number/statistic	completed details	-	-	3	3	9
Business Rules	Do not have business rules	Do not understand the meaning of business rules	Understandable but not reasonable	Do not cover what mentioned in the business domain especially missing core business functionality	Cover all the areas mentioned in the business domain but still have minor errors	Complete and correct	5	3	15
ERD - Entity	Do not have entity	Number of entities and attributes are less than half of the requirement	Number of entities and attributes do NOT meet the requirement and contains errors	Number of entities and attributes meet the requirement but there are several mistakes	Correct but number of entities and attributes are less than the requirement or missing important business functionality	Complete and correct	5	3	15
ERD - Relationship	Do not have relationship	Using incorrect symbol to represent relationship	Missing relationship's name, or not understandable relationship	Do not include cardinality constraints, some major mistakes in ERD	Minor mistakes in the ERD	Complete and correct	5	2	10
Submission	Do not follow the submission guideline	Follow the submission guideline	-	-	-	-	1	1	1

Late submission will NOT be accepted.